- The right-hand side of (44) comes from (30). By definition, $n_{\phi,i}$ is a node of type AND-k, and let $\gamma \triangleq n_{n_{\phi,i},k+1}$ and $\xi_1, \cdots, \xi_{k-1}$ denote the other neighbors of $n_{\phi,i}$. By definition, we have $\gamma = \phi \wedge \xi_1 \wedge \cdots \wedge \xi_{k-1}$, $\tilde{U}_{L,U,n_{\phi,i},j_{\phi,i}} = \min\left(1, U\left(\gamma\right) + \left(1 - L\left(\xi_1\right)\right) + \cdots + \left(1 - L\left(\xi_{k-1}\right)\right)\right)$, $p\left(S_\gamma\right) \leq U\left(\gamma\right)$, and $p\left(S_{\xi_j}\right) \geq L\left(\xi_j\right)$. It is straightforward to verify that
$$S_\gamma = S_\phi \setminus S_{\neg\xi_1} \setminus \cdots \setminus S_{\neg\xi_{k-1}} \tag{49}$$
Therefore,
$$S_\phi \subseteq S_\gamma \cup S_{\neg\xi_1} \cup \cdots \cup S_{\neg\xi_{k-1}} \tag{50}$$
Therefore,
$$\begin{aligned} p\left(S_\phi\right) &\leq p\left(S_\gamma\right) + p\left(S_{\neg\xi_1}\right) + \cdots + p\left(S_{\neg\xi_{k-1}}\right) \\ &= p\left(S_\gamma\right) + \left(1 - p\left(S_{\xi_1}\right)\right) + \cdots + \left(1 - p\left(S_{\xi_{k-1}}\right)\right) \\ &\leq U\left(\gamma\right) + \left(1 - L\left(\xi_1\right)\right) + \cdots + \left(1 - L\left(\xi_{k-1}\right)\right) \end{aligned} \tag{51}$$
Since $p\left(S_\phi\right) \leq 1$, it must be true that
$$p\left(S_\phi\right) \leq \min\left(1, U\left(\gamma\right) + \left(1 - L\left(\xi_1\right)\right) + \cdots + \left(1 - L\left(\xi_{k-1}\right)\right)\right) = \tilde{U}_{L,U,n_{\phi,i},j_{\phi,i}} \tag{52}$$
Hence (44) is true in this scenario.

- The right-hand side of (45) comes from (31). By definition, $n_{\phi,i}$ is a node of type AND-k, and let $\xi_1, \cdots, \xi_k$ denote the other neighbors of $n_{\phi,i}$. By definition, we have $\phi = \xi_1 \wedge \cdots \wedge \xi_k$, $\tilde{L}_{L,U,n_{\phi,i},j_{\phi,i}} = \max\left(0, 1 - \left(1 - L\left(\xi_1\right)\right) - \cdots - \left(1 - L\left(\xi_k\right)\right)\right)$, and $p\left(S_{\xi_j}\right) \geq L\left(\xi_j\right)$. It is straightforward to verify that
$$S_{\neg\phi} = S_{\neg\xi_1} \cup \cdots \cup S_{\neg\xi_k} \tag{53}$$
Therefore,
$$\begin{aligned} p\left(S_{\neg\phi}\right) &\leq p\left(S_{\neg\xi_1}\right) + \cdots + p\left(S_{\neg\xi_k}\right) \\ &\leq \left(1 - L\left(\xi_1\right)\right) + \cdots + \left(1 - L\left(\xi_k\right)\right) \end{aligned} \tag{54}$$
Therefore,
$$p\left(S_\phi\right) = 1 - p\left(S_{\neg\phi}\right) \geq 1 - \left(1 - L\left(\xi_1\right)\right) - \cdots - \left(1 - L\left(\xi_k\right)\right) \tag{55}$$
Since $p\left(S_\phi\right) \geq 0$, it must be true that
$$p\left(S_\phi\right) \geq \max\left(0, 1 - \left(1 - L\left(\xi_1\right)\right) - \cdots - \left(1 - L\left(\xi_k\right)\right)\right) = \tilde{L}_{L,U,n_{\phi,i},j_{\phi,i}} \tag{56}$$
Hence (45) is true in this scenario.

- The right-hand side of (44) comes from (32). By definition, $n_{\phi,i}$ is a node of type AND-k, and let $\xi_1, \cdots, \xi_k$ denote the other neighbors of $n_{\phi,i}$. By definition, we have $\phi = \xi_1 \wedge \cdots \wedge \xi_k$, $\tilde{U}_{L,U,n_{\phi,i},j_{\phi,i}} = \min_{j=1}^{k} U\left(\xi_j\right)$, and $p\left(S_{\xi_j}\right) \leq U\left(\xi_j\right)$. It is straightforward to verify that
$$S_\phi = S_{\xi_1} \cap \cdots \cap S_{\xi_k} \tag{57}$$
Therefore,
$$p\left(S_\phi\right) \leq \min_{j=1}^{k} p\left(S_{\xi_j}\right) \leq \min_{j=1}^{k} U\left(\xi_j\right) = \tilde{U}_{L,U,n_{\phi,i},j_{\phi,i}} \tag{58}$$
Hence (44) is true in this scenario.

- The right-hand side of (45) comes from (33). By definition, $n_{\phi,i}$ is a node of type OR-k, and let $\gamma \triangleq n_{n_{\phi,i},k+1}$ and $\xi_1, \cdots, \xi_{k-1}$ denote the other neighbors of $n_{\phi,i}$. By definition, we have $\gamma = \phi \vee \xi_1 \vee \cdots \vee \xi_{k-1}$, $\tilde{L}_{L,U,n_{\phi,i},j_{\phi,i}} = \max\left(0, L\left(\gamma\right) - U\left(\xi_1\right) - \cdots - U\left(\xi_{k-1}\right)\right)$, $p\left(S_\gamma\right) \geq L\left(\gamma\right)$, and $p\left(S_{\xi_j}\right) \leq U\left(\xi_j\right)$. It is straightforward to verify that
$$S_{\neg\gamma} = S_{\neg\phi} \setminus S_{\xi_1} \setminus \cdots \setminus S_{\xi_{k-1}} \tag{59}$$
Therefore,
$$S_{\neg\phi} \subseteq S_{\neg\gamma} \cup S_{\xi_1} \cup \cdots \cup S_{\xi_{k-1}} \tag{60}$$

Therefore,

$$\begin{aligned} p\left(S_{\neg\phi}\right) &\le p\left(S_{\neg\gamma}\right) + p\left(S_{\xi_1}\right) + \cdots + p\left(S_{\xi_{k-1}}\right) \\ &\le 1 - L\left(\gamma\right) + U\left(\xi_1\right) + \cdots + U\left(\xi_{k-1}\right) \end{aligned} \tag{61}$$

Therefore,

$$p\left(S_\phi\right) = 1 - p\left(S_{\neg\phi}\right) \ge L\left(\gamma\right) - U\left(\xi_1\right) - \cdots - U\left(\xi_{k-1}\right) \tag{62}$$

Since $p\left(S_\phi\right) \ge 0$, it must be true that

$$p\left(S_\phi\right) \ge \max\left(0, L\left(\gamma\right) - U\left(\xi_1\right) - \cdots - U\left(\xi_{k-1}\right)\right) = \tilde{L}_{L,U,n_{\phi,i},j_{\phi,i}} \tag{63}$$

Hence (45) is true in this scenario.

- The right-hand side of (44) comes from (34). By definition, $n_{\phi,i}$ is a node of type OR-k, and let $\gamma \triangleq n_{n_{\phi,i},k+1}$ and $\xi_1, \cdots, \xi_{k-1}$ denote the other neighbors of $n_{\phi,i}$. By definition, we have $\gamma = \phi \vee \xi_1 \vee \cdots \vee \xi_{k-1}$, $\tilde{U}_{L,U,n_{\phi,i},j_{\phi,i}} = U\left(\gamma\right)$, and $p\left(S_\gamma\right) \le U\left(\gamma\right)$. It is straightforward to verify that $S_\phi \subseteq S_\gamma$. Therefore,

$$p\left(S_\phi\right) \le p\left(S_\gamma\right) \le U\left(\gamma\right) = \tilde{U}_{L,U,n_{\phi,i},j_{\phi,i}} \tag{64}$$

Hence (44) is true in this scenario.

- The right-hand side of (45) comes from (35). By definition, $n_{\phi,i}$ is a node of type OR-k, and let $\xi_1, \cdots, \xi_k$ denote the other neighbors of $n_{\phi,i}$. By definition, we have $\phi = \xi_1 \vee \cdots \vee \xi_k$, $\tilde{L}_{L,U,n_{\phi,i},j_{\phi,i}} = \max_{j=1}^{k} L\left(\xi_j\right)$, and $p\left(S_{\xi_j}\right) \ge L\left(\xi_j\right)$. It is straightforward to verify that

$$S_\phi = S_{\xi_1} \cup \cdots \cup S_{\xi_k} \tag{65}$$

Therefore,

$$p\left(S_\phi\right) \ge \max_{j=1}^{k} p\left(S_{\xi_j}\right) \ge \max_{j=1}^{k} L\left(\xi_j\right) = \tilde{L}_{L,U,n_{\phi,i},j_{\phi,i}} \tag{66}$$

Hence (45) is true in this scenario.

- The right-hand side of (44) comes from (36). By definition, $n_{\phi,i}$ is a node of type OR-k, and let $\xi_1, \cdots, \xi_k$ denote the other neighbors of $n_{\phi,i}$. By definition, we have $\phi = \xi_1 \vee \cdots \vee \xi_k$, $\tilde{U}_{L,U,n_{\phi,i},j_{\phi,i}} = \min\left(1, U\left(\xi_1\right) + \cdots + U\left(\xi_k\right)\right)$, and $p\left(S_{\xi_j}\right) \le U\left(\xi_j\right)$. It is straightforward to verify that

$$S_\phi = S_{\xi_1} \cup \cdots \cup S_{\xi_k} \tag{67}$$

Therefore,

$$p\left(S_\phi\right) \le p\left(S_{\xi_1}\right) + \cdots + p\left(S_{\xi_k}\right) \le U\left(\xi_1\right) + \cdots + U\left(\xi_k\right) \tag{68}$$

Since $p\left(S_\phi\right) \le 1$, it must be true that

$$p\left(S_\phi\right) \le \min\left(1, U\left(\xi_1\right) + \cdots + U\left(\xi_k\right)\right) = \tilde{U}_{L,U,n_{\phi,i},j_{\phi,i}} \tag{69}$$

Hence (44) is true in this scenario.

- The right-hand side of (45) comes from (37). By definition, $n_{\phi,i}$ is a node of type IMPLIES, and let $\gamma \triangleq n_{n_{\phi,i},3}$ and $\xi \triangleq n_{n_{\phi,i},2}$ denote the other neighbors of $n_{\phi,i}$. By definition, we have $\gamma = \phi \to \xi$, $\tilde{L}_{L,U,n_{\phi,i},j_{\phi,i}} = 1 - U\left(\gamma\right)$, and $p\left(S_\gamma\right) \le U\left(\gamma\right)$. It is straightforward to verify that $S_{\neg\phi} \subseteq S_\gamma$. Therefore,

$$p\left(S_\phi\right) = 1 - p\left(S_{\neg\phi}\right) \ge 1 - p\left(S_\gamma\right) \ge 1 - U\left(\gamma\right) = \tilde{L}_{L,U,n_{\phi,i},j_{\phi,i}} \tag{70}$$

Hence (45) is true in this scenario.

- The right-hand side of (44) comes from (38). By definition, $n_{\phi,i}$ is a node of type IMPLIES, and let $\gamma \triangleq n_{n_{\phi,i},3}$ and $\xi \triangleq n_{n_{\phi,i},2}$ denote the other neighbors of $n_{\phi,i}$. By definition, we have $\gamma = \phi \to \xi$, $\tilde{U}_{L,U,n_{\phi,i},j_{\phi,i}} = \min\left(1, 1 + U\left(\xi\right) - L\left(\gamma\right)\right)$, $p\left(S_\gamma\right) \ge L\left(\gamma\right)$, and $p\left(S_\xi\right) \le U\left(\xi\right)$. It is straightforward to verify that

$$S_{\neg\gamma} = S_\phi \setminus S_\xi \tag{71}$$

Therefore,

$$S_\phi \subseteq S_{\neg\gamma} \cup S_\xi \tag{72}$$

Therefore,

$$p(S_\phi) \leq p(S_{\neg\gamma}) + p(S_\xi) = 1 - p(S_\gamma) + p(S_\xi) \leq 1 - L(\gamma) + U(\xi)$$

Since $p(S_\phi) \leq 1$, it must be true that

$$p(S_\phi) \leq \min(1, 1 - L(\gamma) + U(\xi)) = \tilde{U}_{L,U,n_{\phi,i},j_{\phi,i}} \tag{73}$$

Hence (44) is true in this scenario.

- The right-hand side of (45) comes from (39). By definition, $n_{\phi,i}$ is a node of type IMPLIES, and let $\gamma \triangleq n_{n_{\phi,i},3}$ and $\xi \triangleq n_{n_{\phi,i},1}$ denote the other neighbors of $n_{\phi,i}$. By definition, we have $\gamma = \xi \to \phi$, $\tilde{L}_{L,U,n_{\phi,i},j_{\phi,i}} = \max(0, L(\xi) + L(\gamma) - 1)$, $p(S_\gamma) \geq L(\gamma)$, and $p(S_\xi) \geq L(\xi)$. It is straightforward to verify that

$$S_{\neg\gamma} = S_{\neg\phi} \setminus S_{\neg\xi} \tag{74}$$

Therefore,

$$S_{\neg\phi} \subseteq S_{\neg\xi} \cup S_{\neg\gamma} \tag{75}$$

Therefore,

$$\begin{aligned} p(S_{\neg\phi}) &\leq p(S_{\neg\xi}) + p(S_{\neg\gamma}) \\ &= 1 - p(S_\xi) + 1 - p(S_\gamma) \\ &\leq 2 - L(\xi) - L(\gamma) \end{aligned} \tag{76}$$

Therefore,

$$p(S_\phi) = 1 - p(S_{\neg\phi}) \geq L(\xi) + L(\gamma) - 1 \tag{77}$$

Since $p(S_\phi) \geq 0$, it must be true that

$$p(S_\phi) \geq \max(0, L(\xi) + L(\gamma) - 1) = \tilde{L}_{L,U,n_{\phi,i},j_{\phi,i}} \tag{78}$$

Hence (45) is true in this scenario.

- The right-hand side of (44) comes from (40). By definition, $n_{\phi,i}$ is a node of type IMPLIES, and let $\gamma \triangleq n_{n_{\phi,i},3}$ and $\xi \triangleq n_{n_{\phi,i},1}$ denote the other neighbors of $n_{\phi,i}$. By definition, we have $\gamma = \xi \to \phi$, $\tilde{U}_{L,U,n_{\phi,i},j_{\phi,i}} = U(\gamma)$, and $p(S_\gamma) \leq U(\gamma)$. It is straightforward to verify that $S_\phi \subseteq S_\gamma$. Therefore,

$$p(S_\phi) \leq p(S_\gamma) \leq U(\gamma) = \tilde{U}_{L,U,n_{\phi,i},j_{\phi,i}} \tag{79}$$

Hence (44) is true in this scenario.

- The right-hand side of (45) comes from (41). By definition, $n_{\phi,i}$ is a node of type IMPLIES, and let $\gamma \triangleq n_{n_{\phi,i},2}$ and $\xi \triangleq n_{n_{\phi,i},1}$ denote the other neighbors of $n_{\phi,i}$. By definition, we have $\phi = \xi \to \gamma$, $\tilde{L}_{L,U,n_{\phi,i},j_{\phi,i}} = \max(1 - U(\xi), L(\gamma))$, $p(S_\xi) \leq U(\xi)$, and $p(S_\gamma) \geq L(\gamma)$. It is straightforward to verify that

$$S_\phi = S_{\neg\xi} \cup S_\gamma \tag{80}$$

Therefore,

$$\begin{aligned} p(S_\phi) &\geq \max(p(S_{\neg\xi}), p(S_\gamma)) \\ &= \max(1 - p(S_\xi), p(S_\gamma)) \\ &\geq \max(1 - U(\xi), L(\gamma)) \\ &= \tilde{L}_{L,U,n_{\phi,i},j_{\phi,i}} \end{aligned} \tag{81}$$

Hence (45) is true in this scenario.